RentEase

Comprehensive Platform, Architecture & Product Report

A Full-Stack Subscription Commerce Platform for Flexible Living in Urban Hubs

Version: 2.4.0 (Production Release)
Runtime: Full-Stack React 19 + Express + MongoDB Atlas
Database Cluster: cluster0.kk44seh.mongodb.net (DB: rentease)
Cloud Provider: Vercel Serverless Gateway
Live Products in DB: 22 Active Items
Classification: Production System Specification

1. Executive Summary

RentEase is an end-to-end furniture and smart appliances subscription platform tailored for young professionals, students, interns, and remote-first nomads across tier-1 Indian metros (Bengaluru, Mumbai, Delhi-NCR, Hyderabad, and Pune).

The platform solves the friction of traditional home setup: exorbitant capital expenditure on brand-new furniture, the logistical nightmare of relocation, and rapid depreciation. By decoupling utility from ownership through flexible tenure commitments (3, 6, and 12-month plans) with 100% refundable deposits, comprehensive zero-charge doorstep maintenance, and guaranteed 48-hour delivery, RentEase transforms durable goods into a seamless service.

Key Business & Customer Pillars:

- **Zero Upfront Capital:** Fractional monthly rent starting as low as ₹299/mo instead of ₹30,000+ upfront purchase.
- **Duration-Discounted Curves:** Algorithmic tiered pricing: 12-month contracts yield up to 25% rent savings versus 3-month plans.
- **100% Refundable Deposits:** Security deposits held securely and refunded via digital audit trails within 48h of return.
- **Free Lifetime Servicing:** Complimentary bi-annual deep sanitization, breakdown servicing, and free relocation support.
- **Full-Stack Operations:** Integrated logistics portal for order tracking, tenure extensions, returns, and technician dispatch.

2. System Architecture & Tech Stack

RentEase is engineered with a modern decoupling strategy allowing seamless unified execution on local development and high-availability serverless deployment on Vercel backed by persistent cloud storage.

Technology Matrix:

LAYER	TECHNOLOGY	ROLE & SPECIFICATION	DEPLOYMENT TARGET
Client Frontend	React 19 + TypeScript + Vite 8	Single Page Application with fast HMR, strict types	Vercel Edge
Styling Engine	Tailwind CSS v4 (Modern CSS AST)	Zero-runtime utility engine, bespoke design constitution	Hashed CSS Output
Icons & Assets	Lucide React + WebP Assets	Scalable vector icons & high-fidelity photographic imagery	Bundled
API Layer	Express.js + Node.js 22 LTS	RESTful modular endpoints with CORS & URL normalization	Vercel Serverless
Database Engine	MongoDB Atlas (Cluster 0)	Distributed Document DB with ACID transactions & indexing	AWS Mumbai (ap-south-1)
DevOps Gateway	Vercel URL Rewrite Gateway	Path normalizer routing x-matched-path into serverless core	Vercel Edge Network

3. Cloud Database Data Model (MongoDB Atlas)

- **collection("products"):** Primary rental inventory (title, category, 3/6/12m pricing, deposit, stockCount, condition, warranty).
- **collection("orders"):** Customer rental agreements (orderNumber, line items, delivery address, monthly billing, trackingSteps).
- **collection("tickets"):** Doorstep maintenance tickets (issueCategory, priority, preferredDate, technicianName, resolutionNotes).
- **collection("claims"):** Post-return damage audits & security deposit deduction calculation (claimStatus, damageDeduction, refundAmount).
- **collection("cities"):** Geographic operational hubs (Bengaluru,

Mumbai, Delhi-NCR, activeHubs, pinCodes, delivery SLA).

4. Core Platform Capabilities & Workflows

A. Storefront Catalog & Dynamic Pricing Curves

Customers filter across Furniture, Smart Appliances, and Pre-configured Room Bundles (e.g. Student 1BHK, WFH Executive Suite). Product detail pages compute dynamic monthly pricing based on the chosen tenure (3m, 6m, 12m) and calculate the refundable security deposit in real time.

B. Express Checkout & Inventory Hold

The cart drawer enforces tenure selection per item, applies delivery address checks against active operational hubs (e.g., verifying 560103 in Bengaluru), and transitions orders into "scheduled" delivery with sequential tracking milestones (Confirmed -> Quality Passed -> In Transit -> Setup Done).

C. Self-Service User Rentals Portal

Active subscribers can manage their contracts from their customer account: one-click tenure extensions, scheduling semi-annual professional sanitization, raising maintenance repair requests, requesting free relocation to a new address, or booking return pick-ups.

D. Vendor & Admin Operations Console

A complete internal operations dashboard providing real-time MRR tracking, stock utilization percentage, dispatch queue control, technician assignment for maintenance tickets, damage auditing, and deposit refund authorization.

5. REST API Specifications

METHOD & ENDPOINT	DESCRIPTION	DATA SOURCE
GET /api/health	Cluster connectivity, live database status & inventory count	MongoDB Atlas
GET /api/products	Fetch dynamic catalog with category & search parameters	products collection
POST /api/products	Add new rental item directly to cloud catalog	products collection
GET /api/orders	Retrieve all customer contracts & dispatch status	orders collection
POST /api/orders	Place subscription order & deduct available inventory	orders collection
GET /api/tickets	Fetch ongoing servicing requests with technician assignments	tickets collection
POST /api/tickets	Submit doorstep maintenance request	tickets collection
PUT /api/claims/:id	Authorize deposit refund with itemized damage deductions	claims collection
GET /api/cities	Fetch operational cities & hub service coverage	cities collection

6. Live Production Metrics & Audit Snapshot

TOTAL DYNAMIC PRODUCTS	ACTIVE RENTALS & ORDERS	SERVICE CITIES
22 Items	**2 Contracts**	**5 Hubs**
100% Persisted in Atlas	Bengaluru Bellandur Hub	BLR, MUM, DEL, HYD, PNQ

7. Reliability, Security & Vercel Gateway Hardening

• **Zero Hardcoded Mocks:** The codebase is configured in pure dynamic database mode. If the cloud database returns an empty collection, the system safely renders verified empty states rather than fictitious mockups.

• **Vercel Serverless Path Normalization:** The Express application mounts an intelligent URL normalizer reading the x-matched-path header, completely bypassing edge rewrite truncation and routing POST /api/products flawlessly.

• **Defensive Array Guards:** All API services and UI containers employ Array.isArray() validation preventing TypeError: filter is not a function across all network edge cases.

• **CORS & Header Security:** Full preflight OPTIONS handling and permissive CORS headers allowing cross-origin requests between Vercel microservices and custom frontends.

8. Strategic Growth Roadmap

[Phase 1 (Current)] Live dynamic MongoDB Atlas integration, multi-tenure pricing, self-service customer portal, and comprehensive admin suite.

[Phase 2 (Q2 2027)] Automated KYC verification with DigiLocker API integration, automated recurring mandate billing (e-NACH/UPI Autopay).

[Phase 3 (Q3 2027)] AI-powered computer vision damage assessment during return pick-ups and IoT asset tracking for high-value appliances.

[Phase 4 (Q4 2027)] Tier-2 expansion into Chandigarh, Ahmedabad, Kochi, and Jaipur with franchised local refurbishment warehouses.

PREPARED FOR: RentEase Executive Committee & Engineering Operations
Author: Senior Lead Systems Engineer | Platform Status: 100% Operational | Database: Verified Connected